# Predicting Vegetation Growth Risk for Tree Trimming Prioritization Using Machine Learning

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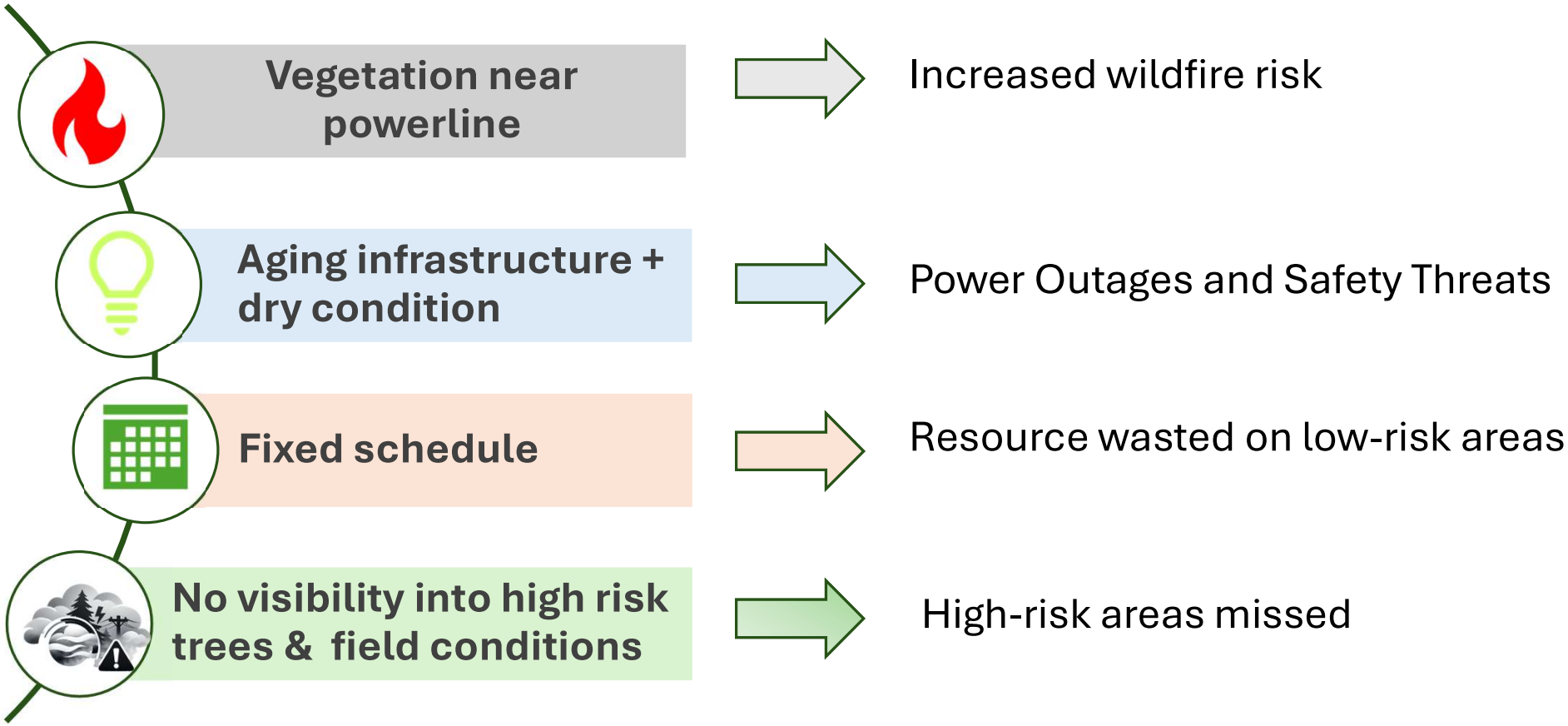
*Billion-dollar events to affect California from 1980 to 2024 (CPI-Adjusted)*

| Disaster Type | Events | Events/Year | Percent Frequency | Total Costs | Percent of Total Costs |
|---|---|---|---|---|---|
| Drought | 14 | 0.3 | 30.4% | $10.0B-$20.0B | 10.7% |
| Flooding | 6 | 0.1 | 13.0% | $10.0B-$20.0B | 12.4% |
| Freeze | 3 | 0.1 | 6.5% | $10.0B-$20.0B | 10.0% |
| Severe Storm | 4 | 0.1 | 8.7% | $2.0B-$5.0B | 2.3% |
| Tropical Cyclone | -- | -- | -- | -- | -- |
| Wildfire | 19 | 0.4 | 41.3% | $100.0B-$200.0B | 64.6% |
| Winter Storm | -- | -- | -- | -- | -- |
| **All Disasters** | **46** | **1.0** | **100.0%** | **$100.0B-$200.0B** | **100.0%** |

Reference:https://www.ncei.noaa.gov/access/billions/state-summary/CA

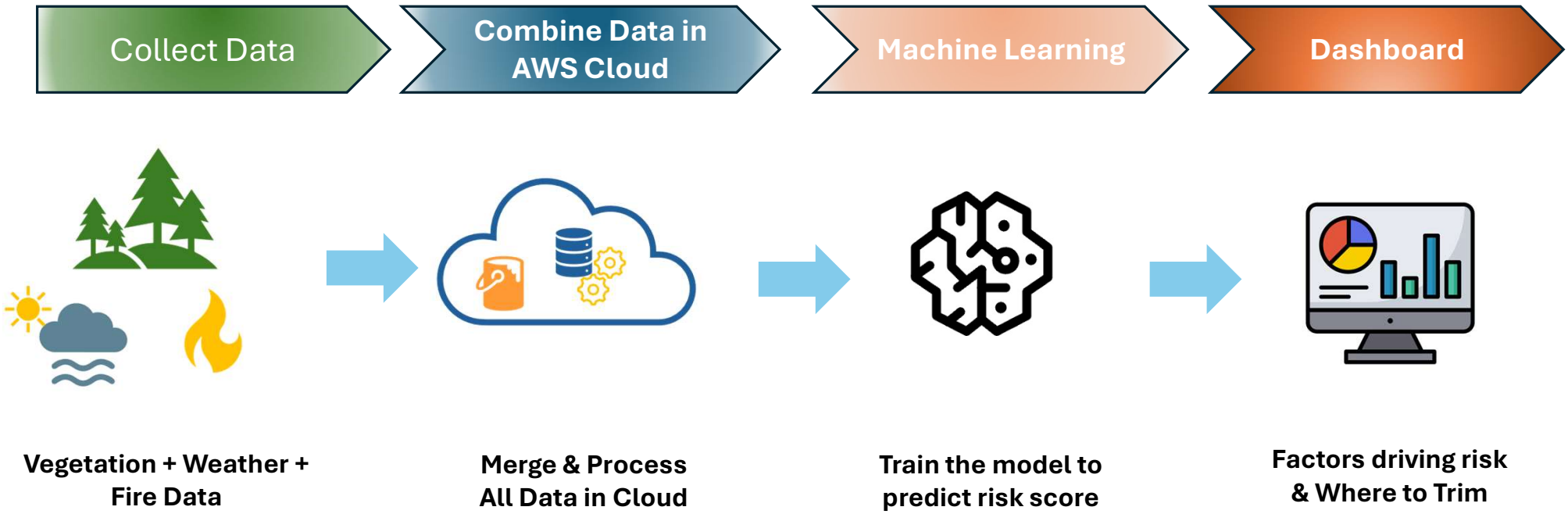
# The Problem


Vegetation near powerline
Increased wildfire risk
Aging infrastructure + dry condition
Power Outages and Safety Threats
Fixed schedule
Resource wasted on low-risk areas
No visibility into high risk trees & field conditions
High-risk areas missed

# WHY TRADITIONAL METHODS ARE NOT ENOUGH

- Manual inspections are **slow** and **costly**
- **Reactive** approach instead of proactive risk management
- Decisions based on **limited data** and human judgment
- **Miss** key risk factors (weather, vegetation, fire history)
- Cannot keep up with **climate change**
- **High** operational **cost** due to inefficient trimming

## OUR SOLUTION



# Key Findings

Hot, dry, and dense vegetation drives higher risk

Past fires strongly predict future risk

High-risk areas are concentrated, not evenly spread

Not all areas are risky ,enabling targeted action

Drier vegetation = higher fire danger (fuel moisture)

# DASHBOARD

- Interactive dashboard to visualize risk levels
- Allows filtering by location and risk category
- Helps decision-makers quickly identify high-risk areas
- Supports data-driven planning and action

# Who Benefits From Our Solution?

| Utilities | Government | Houseowners | Land Mgmt | Insurance |
|---|---|---|---|---|
| Fewer Outages | Early risk detection | Improved safety | Optimized operations | Better risk assessments |
| Improved Customer Satisfaction | Proactive wildfire | Smarter property decisions | Lower operational cost | Faster claims response |

*Delivering Value Across the Ecoystem*

*Machine learning insights enable smarter decisions across all stakeholders, reducing risk, cost, and impact*

# Our Recommendation

| Recommendation | Action | Expected Benefit |
|---|---|---|
| Act on high-priority zones now | Send crews to 81,000 flagged locations | Reduce wildfire ignition risk this season |
| Stop routine trimming in low-risk zones | Redirect crews to high-risk areas | Save 30–40% of trimming costs |
| Adopt dashboard company-wide | Train field supervisors this quarter | Faster, smarter daily decisions |
| Refresh data seasonally | Update weather and fire data every 3 months | Maintain accuracy as conditions change |

# Future Improvements/Next Steps

REAL-TIME WEATHER INTEGRATION

EXPAND DATA SOURCES

AUTOMATED ALERTS